

CSE110 Spring 2024 Lab Exam - 02 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;

public class Vault{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the secret number: ");
        int num = sc.nextInt();

        int vaultValue = 0;
        while (num > 0) {
            int digit = num % 10;

            if (digit % 2 == 0) {
                vaultValue += digit * digit;
            }
            else if (digit % 2 != 0 && digit < 5) {
                vaultValue -= digit * digit * digit;
            }

            else if (digit % 3 == 0) {
                vaultValue += digit;
            }
            else{
                vaultValue -=digit;
            }

            num /= 10;
        }

        System.out.println("Vault Value = " + vaultValue);
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Proper import and java syntax	1
2	Creating scanner object and taking inputs and prompts properly	1
3	For loop implementation and extract each digit from given number properly	3
4	Checking conditions and calculate the value	4
5	Print the output properly	1

[SET-B]

SOLUTION

```
import java.util.Scanner;

public class chamber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the secret number: ");
        int num = sc.nextInt();

        int vaultValue = 0;
        while (num > 0) {
            int digit = num % 10;

            if (digit % 2 == 0) {
                vaultValue -= digit * digit;
            }
            else if (digit % 2 != 0 && digit <= 3 ) {
                vaultValue += digit * digit * digit;
            }

            else if (digit % 3 == 0) {
                vaultValue += digit;
            }
            else{
                vaultValue -=digit;
            }

            num /= 10;
        }

        System.out.println("Vault Value = " + vaultValue);
    }
}
```

[CSE110 Spring 2024 Lab Exam - 0X Tentative Solutions and Rubrics](#)

RUBRIC

SL	Points To Meet	Marks [10]
1	Proper import and java syntax	1
2	Creating scanner object and taking inputs and prompts properly	1
3	For loop implementation and extract each digit from given number properly	3
4	Checking conditions and calculate the value	4
5	Print the output properly	1